



IEEE INTERNATIONAL CONFERENCE
ON ROBOTICS AND AUTOMATION

2027
SEOUL



Beyond Policy Alignment

Closing the Planning-Learning Loop for Robot Control with Learned World Models

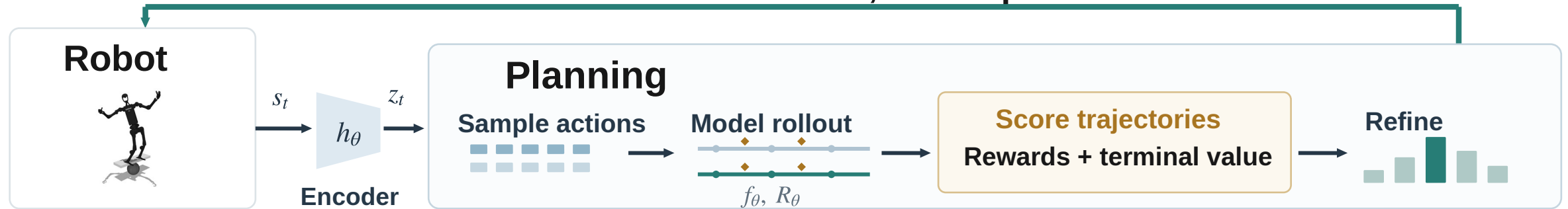
Anonymous Authors

<https://icra27-pl-mpc.github.io/>

TD-MPC: planning with a learned world model

Model predictive control: optimize a short action sequence.

Execute one action, then replan



World model

Predicts action outcomes

Critic

Estimates future return

Policy

Proposes candidate actions

Execute → collect experience → learn → replan

Studied problem: beyond policy alignment

Policy alignment trains the policy toward planner behavior.

1 Critic supervision

Later rewards reach earlier values through repeated updates.

2 Terminal-value evaluation

Uncertain values can change trajectory ranking.

3 Planner-to-policy transfer

Planner experience varies in quality.

PL-MPC addresses all three interfaces.

Closing the planning-learning loop

PL-MPC · Planning-Learning MPC

Studied Problem

Policy alignment addresses only one part of the planning-learning loop.

Our Solution

Unchanged world-model architecture and MPPI optimizer.

1. MTD · Hybrid multi-step TD targets

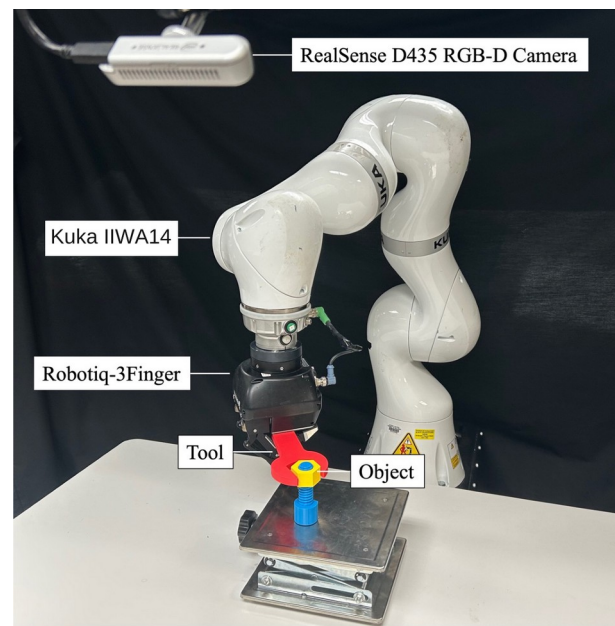
More observed rewards before bootstrapping.

2. ATE · Adaptive terminal estimates

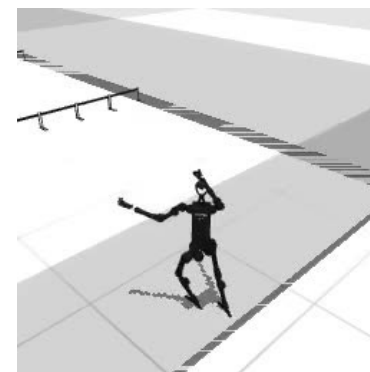
Penalize terminal values using critic disagreement.

3. RAD · Return-weighted actor distillation

Weight imitation by realized episode return.



balance-hard



hurdle

Humanoid control and zero-shot sim-to-real transfer

Result balance-hard: 98 → 387 TAR

hurdle: 199 → 466 TAR

Robot: 61.3% → 74.2%

vs. TD-M(PC)² · TAR: Total Average Return · Robot: observed success on training size (31 trials/method).

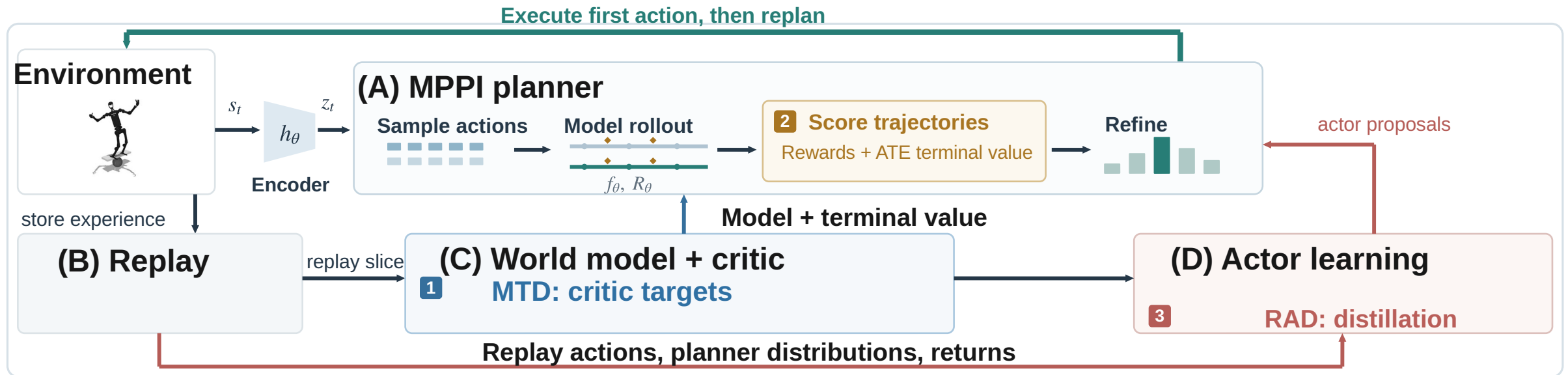
Related work

Policy learning, value learning, and terminal-value evaluation

Method	Policy learning	TD targets	Terminal value
TD-M(PC) ²	Planner constraint	One-step TD	Q bootstrap
BMPC	Expert imitation + lazy reanalysis	One-step TD (model-based)	Value bootstrap
BOOM	Q-weighted imitation	One-step TD	Q bootstrap
PO-MPC	KL-regularized RL + adaptive prior	One-step TD	Q bootstrap
PL-MPC (Ours)	Constraint + episode-return weighting	Hybrid multi-step TD (replay + model)	Q bootstrap – disagreement penalty

PL-MPC architecture

Plan → execute → learn → replan



1 Critic supervision

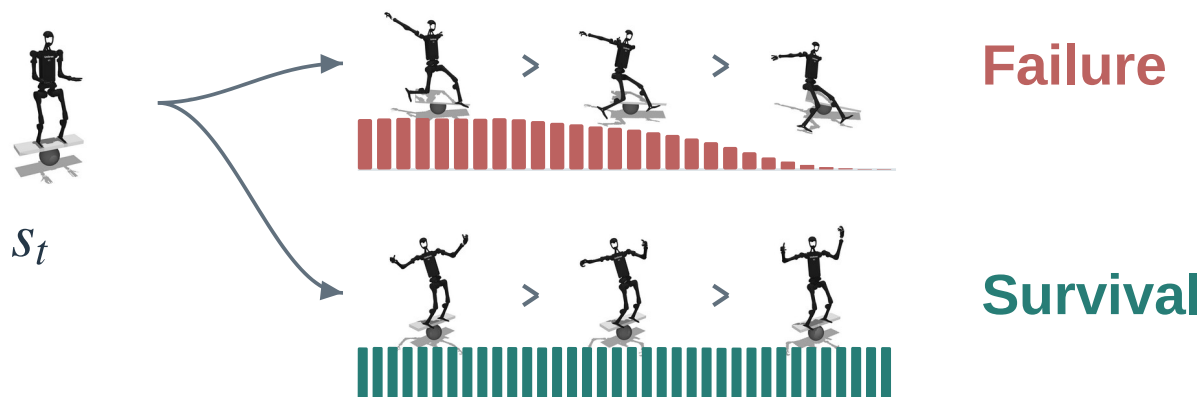
2 Terminal-value evaluation

3 Planner-to-policy transfer

1. Hybrid multi-step TD targets (MTD)

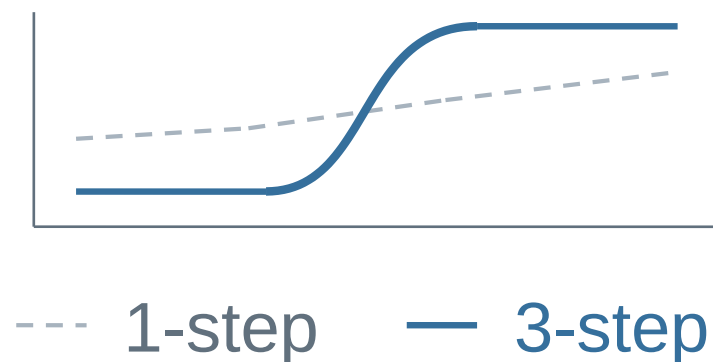
Later rewards distinguish falling from surviving sooner.

Same initial state Illustrative trajectory pair



Illustrative realized rewards

Fall-survival
boundary



Hybrid targets $n = H = 3$

Observed / model reward counts

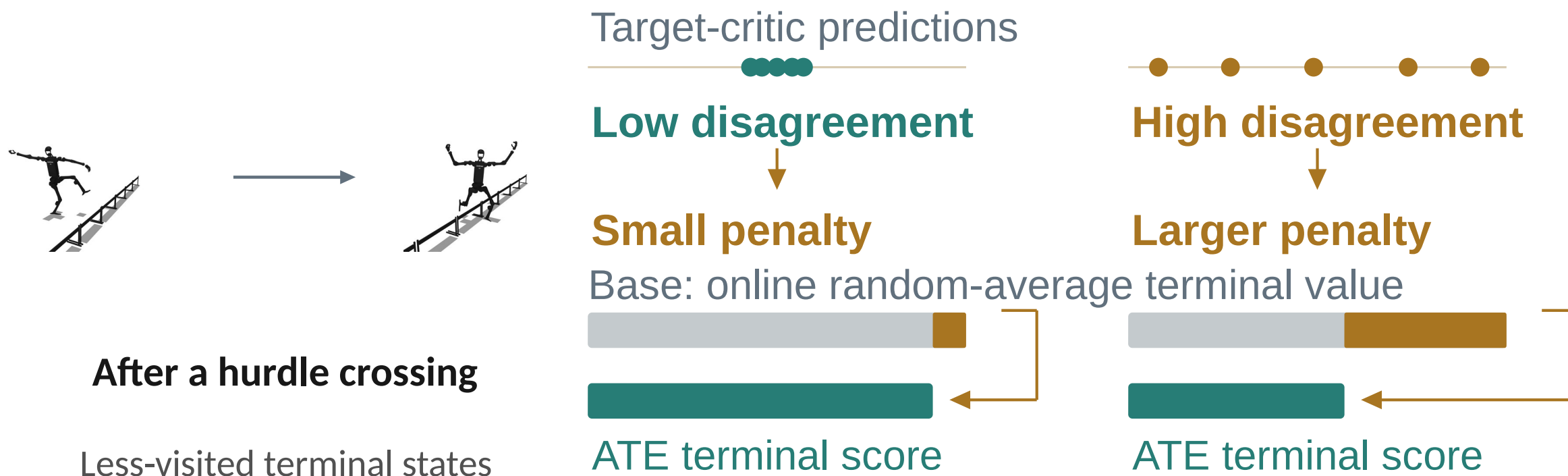
✗	$y_0^{(3)}$	(3, 0)	r_0	r_1	r_2
✓	$y_1^{(3)}$	(2, 1)	r_1	r_2	$\tilde{r}_3$
	$y_2^{(3)}$	(1, 2)	r_2	$\tilde{r}_3$	$\tilde{r}_4$

Actor bootstrap at each endpoint

Observed rewards +
model-predicted tail

2. Adaptive terminal estimates (ATE)

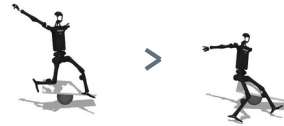
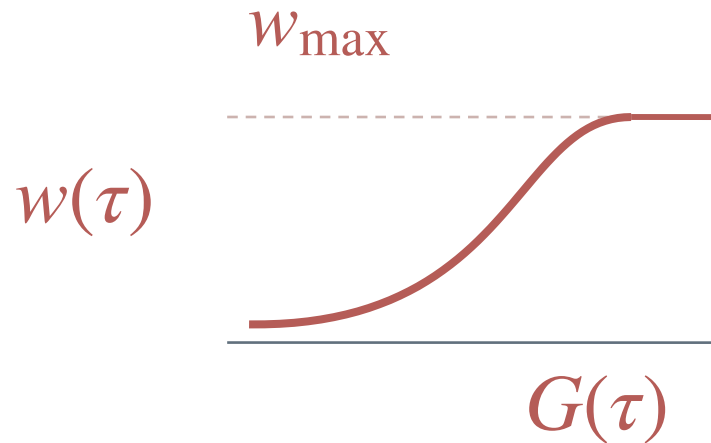
Reduce reliance on uncertain terminal values after hurdle crossing.



3. Return-weighted actor distillation (RAD)

Realized episode return determines imitation weight.

Higher episode return
→ stronger imitation

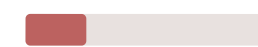


Planner-executed replay actions

Lower-return episode

$G(\tau)$

small $w(\tau)$



Higher-return episode

$G(\tau)$

large $w(\tau)$



Loss after warmup

Weighted imitation
of planner actions

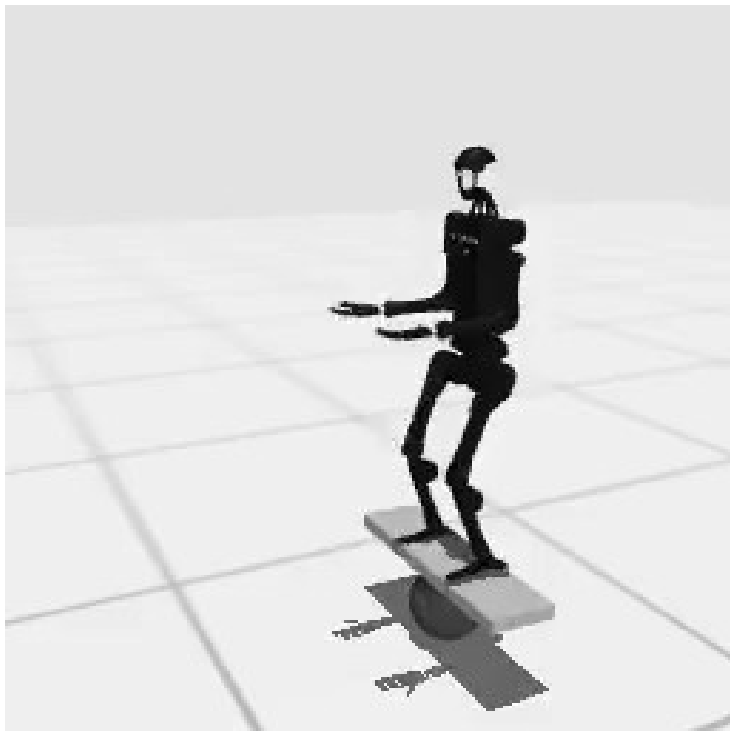
Warmup: collect useful planner trajectories → activate RAD

HumanoidBench: balance-hard

5× speed

Maintaining balance for the full episode

PL-MPC

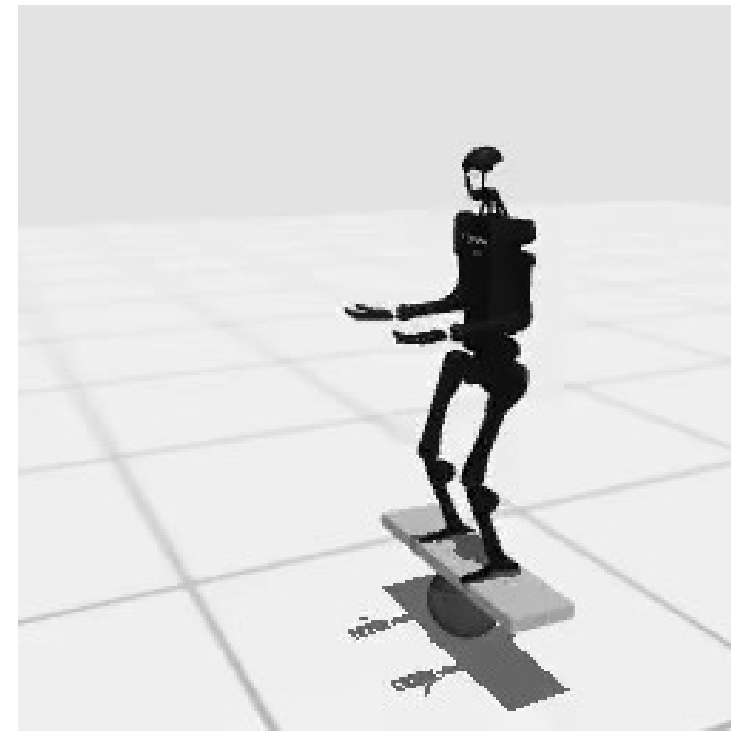


Full episode

Best-seed success: 50.9%

TAR: 387 ± 255

TD-M(PC)²



Falls

Best-seed success: 0%

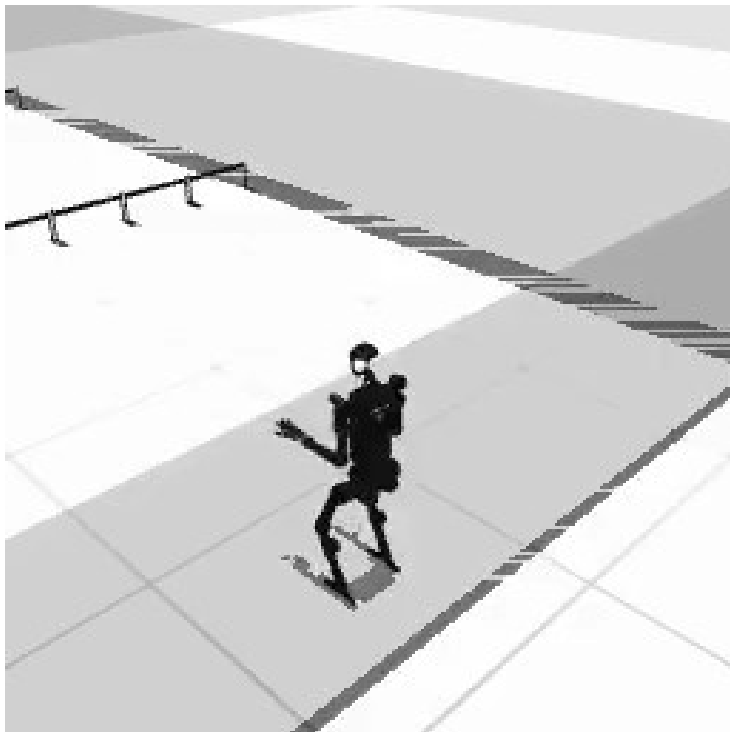
TAR: 98 ± 18

HumanoidBench: hurdle

5× speed

Crossing successive hurdles

PL-MPC

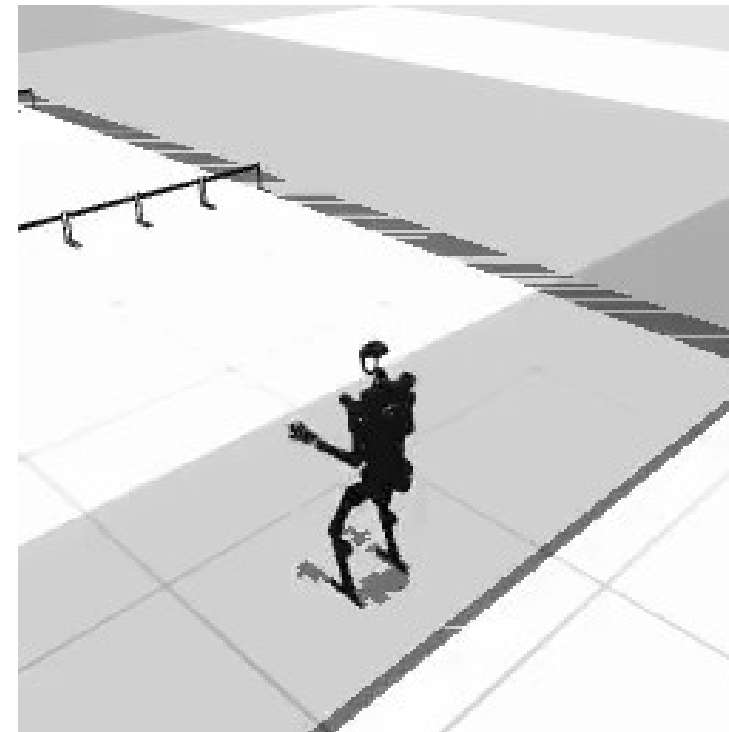


Full episode

Best-seed success: 58.0%

TAR: 466 ± 200

TD-M(PC)²



Falls

Best-seed success: 0%

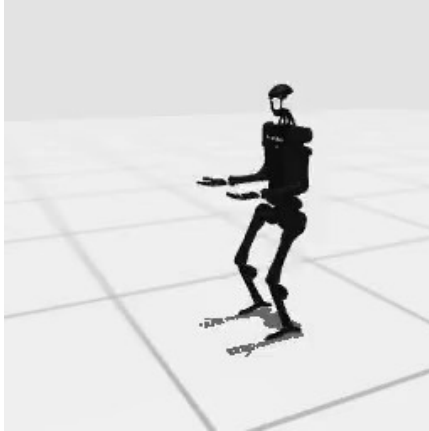
TAR: 199 ± 13

Other HumanoidBench tasks

7× speed

Competent performance on other tasks

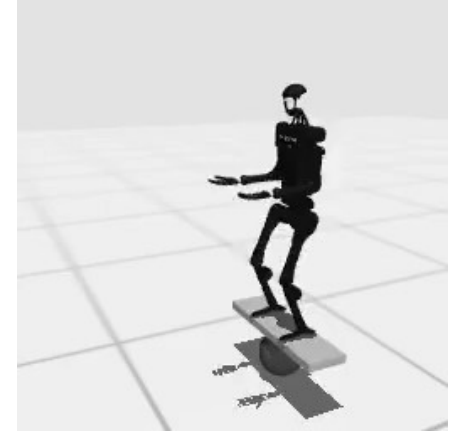
run



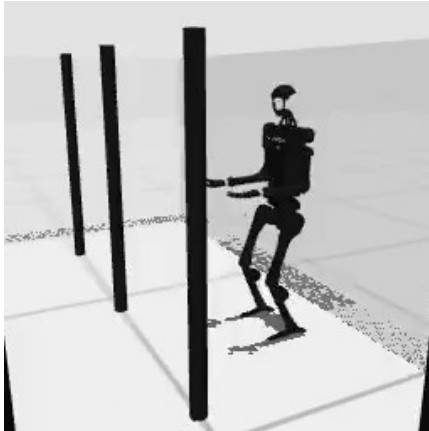
crawl



balance-simple



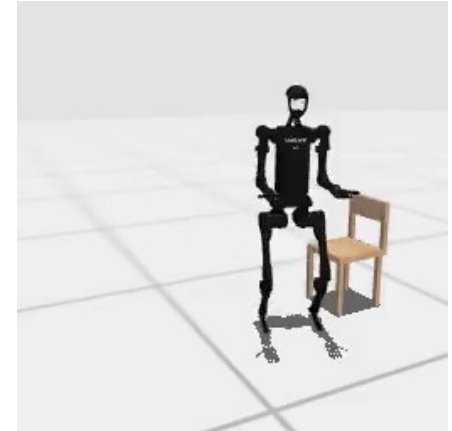
pole



slide



sit-hard



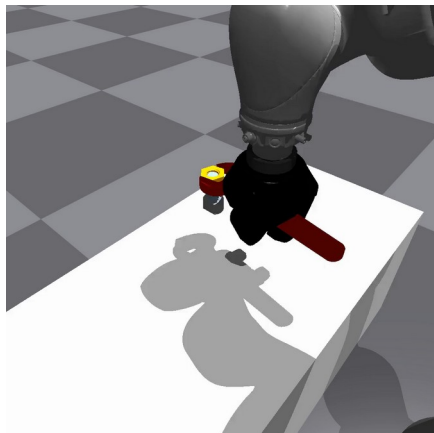
Wrench-nut alignment in simulation

3× speed

Isaac Gym · Training size 5 · Three starting configurations

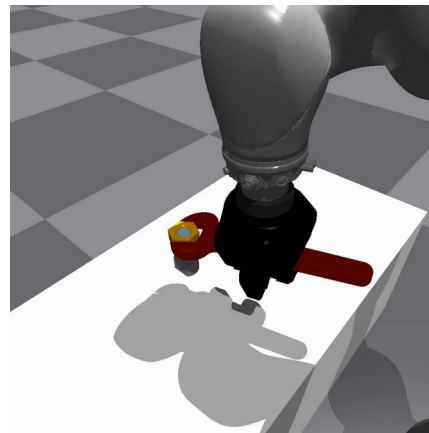
PL-MPC

Scene 1



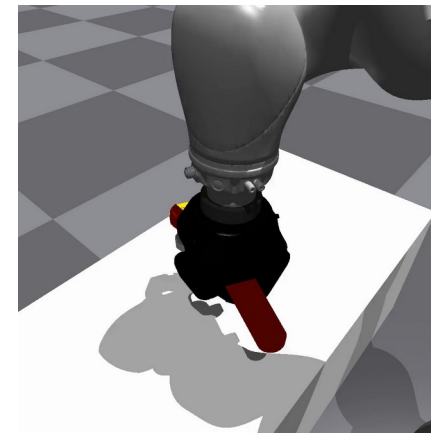
Success

Scene 3



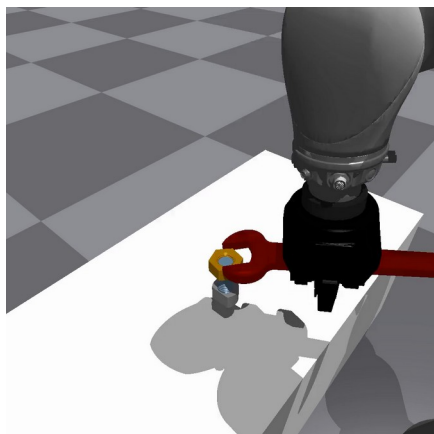
Success

Scene 4

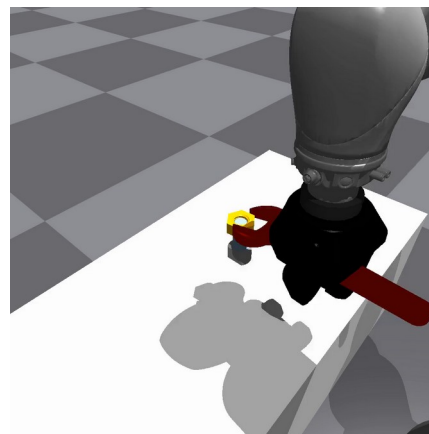


Success

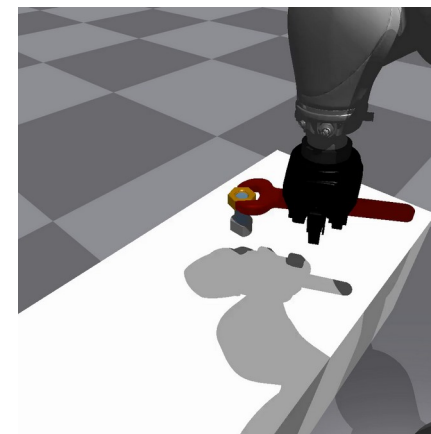
TD-M(PC)²



Timeout



Timeout



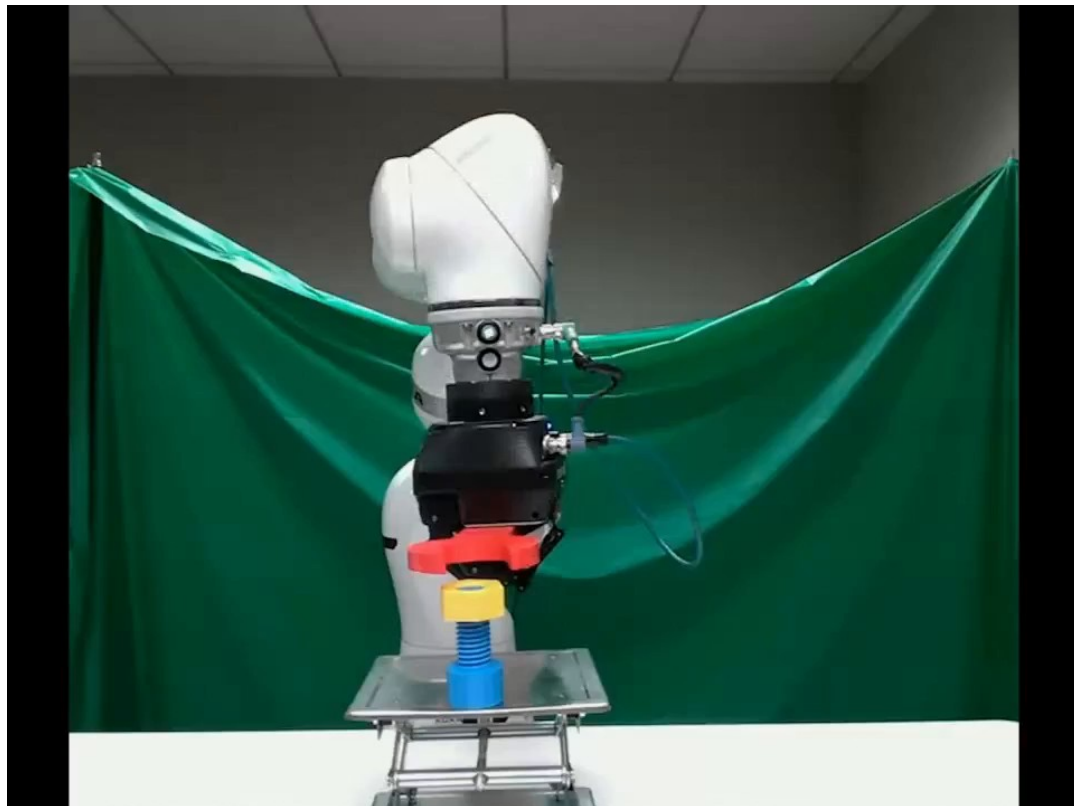
Success

Zero-shot wrench-nut alignment

4× speed

Simulation-trained world model • Online planning on the real robot

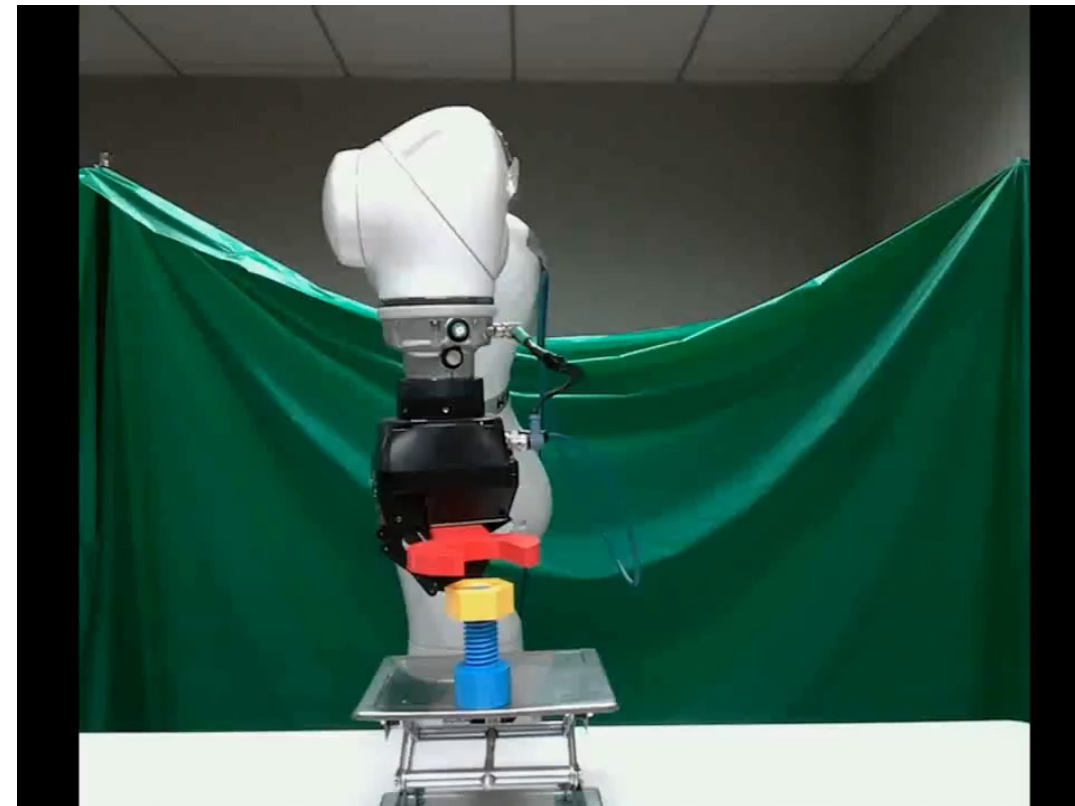
PL-MPC



Success

74.2% • 23/31

TD-M(PC)²



Timeout

61.3% • 19/31

Additional real-robot demonstrations

4× speed

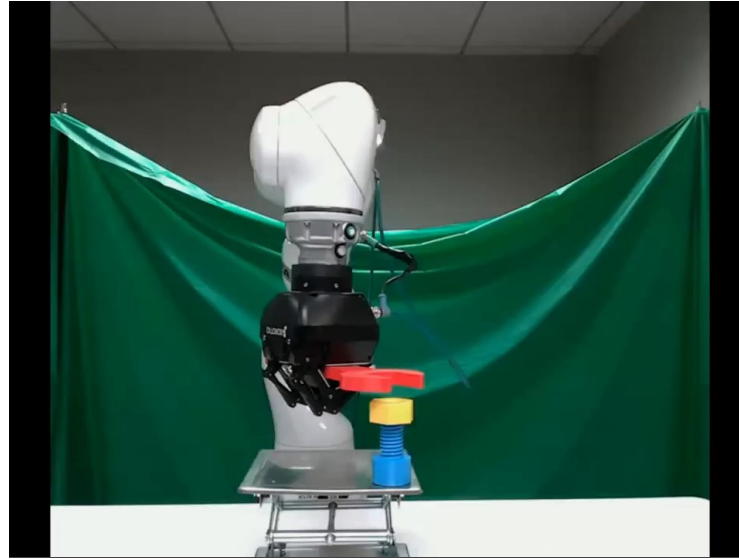
PL-MPC · Zero-shot sim-to-real · No real-world fine-tuning

Training size 5 · 46 mm



Success

Training size 5 · 46 mm



Success

Unseen size 3 · 36 mm



Success

Transfer to unseen object sizes

6× speed

Size 3 · 36 mm

PL-MPC



Success

80.0%

12/15 trials

TD-M(PC)²



Timeout

73.3%

11/15 trials

Size 1 · 30 mm

PL-MPC



Success

33.3%

5/15 trials

TD-M(PC)²



Success

20.0%

3/15 trials

Summary: closing the planning-learning loop

PL-MPC • Planning-Learning MPC

Studied Problem

Policy alignment addresses only one part of the planning-learning loop.

Our Solution

PL-MPC • A policy-constrained TD-MPC extension.

1. MTD • Hybrid multi-step TD targets

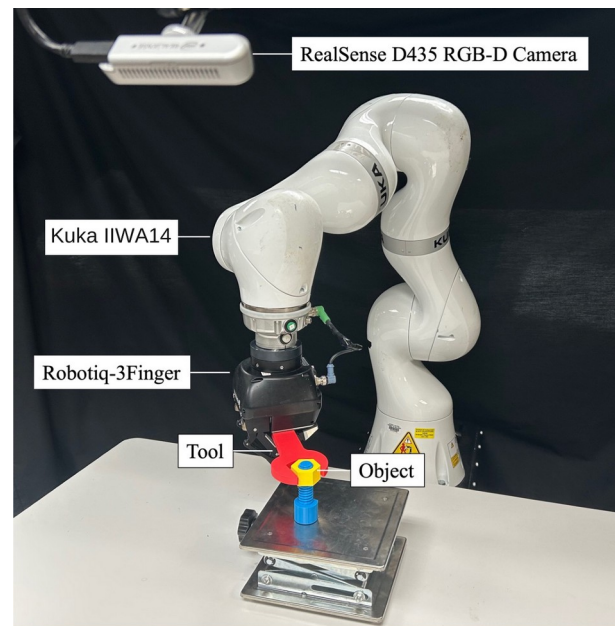
More observed rewards before bootstrapping.

2. ATE • Adaptive terminal estimates

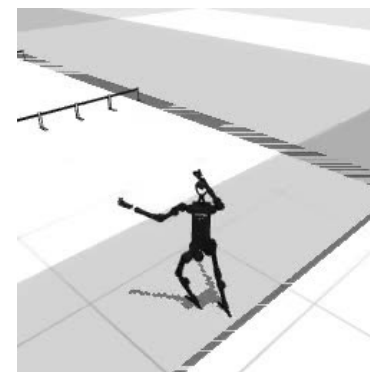
Penalize terminal values using critic disagreement.

3. RAD • Return-weighted actor distillation

Weight imitation by realized episode return.



balance-hard



hurdle

<https://icra27-pl-mpc.github.io/>

Result balance-hard: 98 → 387 TAR

hurdle: 199 → 466 TAR

Robot: 61.3% → 74.2%

vs. TD-M(PC)² • TAR: Total Average Return • Robot: observed success on training size (31 trials/method).